

1. Initial Kernel Parameter Estimation Procedure

Inputs

- Data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$.
- Estimation type: 'cov' or 'scale'.

Outputs

- Normalization factor $\eta \in \mathbb{R}$.
- Inverse matrix $\Psi^{-1} \in \mathbb{R}^{D \times D}$.

Procedure A: Covariance-based Estimation ('cov')

1. **Empirical Covariance Matrix (Σ):** Let $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$.

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^N (\mathbf{x}_i - \hat{\mu})^T (\mathbf{x}_i - \hat{\mu})$$

2. **Covariance Matrix Validation:** Check if $\det(\Sigma)$ is non-negligible and positive.
3. **Normalization Factor (η):**

$$\eta = \frac{1}{(2\pi)^{D/2} \sqrt{\det(\Sigma)}}$$

4. **Inverse Covariance Matrix ($\Psi^{-1} = \Sigma^{-1}$):** $\Psi^{-1} = \Sigma^{-1}$.

Procedure B: Scale-based Estimation ('scale')

1. **Inverse Matrix ($\Psi^{-1} = \mathbf{I}$):** $\Psi^{-1} = \mathbf{I}$ (identity matrix).
2. **Normalization Factor (η):**

$$\eta = \frac{1}{(2\pi)^{D/2}}$$

2. Parameterized Kernel Function

Inputs

- Data matrix $\mathbf{X} \in \mathbb{R}^{N_X \times D}$.
- Data matrix $\mathbf{Y} \in \mathbb{R}^{N_Y \times D}$.

- Initial normalization factor $\eta_0 \in \mathbb{R}$.
- Initial inverse matrix $\Psi_0^{-1} \in \mathbb{R}^{D \times D}$.
- Bandwidth hyperparameter $h \in \mathbb{R}$, $h \neq 0$.

Output

- Normalized kernel matrix $\mathbf{K} \in \mathbb{R}^{N_X \times N_Y}$.

Calculation Steps

1. Pre-computation:

- Scaled Inverse Matrix: $\Psi_h^{-1} = \frac{\Psi_0^{-1}}{h^2}$.
- Scaled Normalization Factor: $\eta_h = \frac{\eta_0}{h^D}$ (with boundary conditions for $\eta_0 = 0$ or very small h^D).

2. Quadratic Form Matrix (Q): $Q_{ij} = (\mathbf{x}_i - \mathbf{y}_j)^T \Psi_h^{-1} (\mathbf{x}_i - \mathbf{y}_j)$. Expanded as:

$$Q_{ij} = (\text{diag}(\mathbf{X} \Psi_h^{-1} \mathbf{X}^T))_i - 2(\mathbf{X} \Psi_h^{-1} \mathbf{Y}^T)_{ij} + (\text{diag}(\mathbf{Y} \Psi_h^{-1} \mathbf{Y}^T))_j$$

3. Unnormalized Kernel-like Values (P):

$$P_{ij} = \eta_h \exp \left(-\frac{1}{2} Q_{ij} \right)$$

4. Final Normalization: Let $P_{max} = \max_{i,j} (P_{ij})$.

$$K_{ij} = \begin{cases} 0 & \text{if } P_{max} \approx 0 \\ \frac{P_{ij}}{P_{max}} & \text{otherwise} \end{cases}$$